

Worksheet 10

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Subject Name: ADBMS

Subject Code: 23CSP-333

1. Aim: Perform CRUD Operations using No-SQL Database: MongoDB

2. Objective:

- Create and manage databases and collections.
- Perform **CRUD operations** — Create, Read, Update, and Delete documents.
- Use **aggregation** for grouping and summarizing data.
- Understand how to modify, query, and manage collections efficiently in MongoDB.
- Gain practical experience with real-world data handling through command-line operations.

3. Code:

```
// 1. Show all databases show
```

```
use
```

```
// 2. Create or switch to your database use
```

```
use shivanshu
```

```
// 3. Check current database db
```

```
// 4. Create a collection (optional)
```

```
db.createCollection("movies") //
```

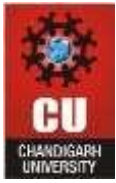
5. Insert Operations

```
db.movies.insertOne({ title:
```

```
"Inception", director:
```

```
"Christopher Nolan", year: 2010,
```

```
rating: 8.8,
```



genre: "Sci-Fi",

features: ["Mind-bending", "Action"], added_by:

"Shivanshu"

})

db.movies.insertMany([

{

title: "Interstellar",

director: "Christopher Nolan",

year: 2014, rating: 8.6, genre:

"Sci-Fi", features: ["Space",

"Drama"], added_by:

"Shivanshu"

},

{

title: "Avatar",

director: "James Cameron",

year: 2009,

rating: 7.8,

genre: "Adventure", features:

["3D", "Action"], added_by:

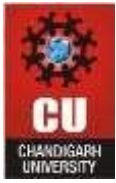
"Shivanshu"

},

{ title: "The Dark

Knight", director:

"Christopher Nolan",



year: 2008, rating: 9.0,

genre: "Action",

features: ["Batman",

"Crime"], added_by:

"Shivanshu"

}

])

// 6. Show all collections show collections // 7.

Read Operations db.movies.find().pretty()

db.movies.findOne() db.movies.find({}, { title:

1, year: 1, _id: 0 }) db.movies.find({ genre:

"Action" }) db.movies.find({ "details.language":

"English" })

// 8. Update Operations

db.movies.updateOne(

{ title: "Avatar" },

{ \$set: { rating: 8.1, genre: "Sci-Fi Adventure" } }

)

db.movies.updateOne(

{ title: "Inception" },

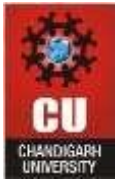
{ \$push: { features: "Thriller" } }

)

db.movies.updateOne(

{ title: "Inception" },

{ \$pull: { features: "Action" } }



)

```
db.movies.updateMany(
```

```
  { director: "Christopher Nolan" },
```

```
  { $set: { language: "English" } }
```

)

```
db.movies.updateMany({}, { $unset: { added_by: "" } })
```

```
db.movies.updateMany({}, { $set: { color: "Full HD" } })
```

```
db.movies.updateOne(
```

```
  { title: "Tenet" },
```

```
  { $set: { director: "Christopher Nolan", year: 2020, rating: 7.5 } },
```

```
  { upsert: true } 
```

)

```
// 9. Delete Operations db.movies.deleteOne({ title:
```

```
"Avatar" }) db.movies.deleteMany({ director:
```

```
"Christopher Nolan" }) db.movies.deleteMany({}) // 10.
```

```
Grouping (Aggregation) db.movies.aggregate([
```

```
  { $group: { _id: "$director", total_movies: { $sum: 1 } } }
```

])

```
db.movies.aggregate([
```



```
{ $group: { _id: "$genre", total_movies: { $sum: 1 } } }
```

```
]
```

```
// 11. Drop collection
```

```
db.movies.drop() //
```

```
12. Drop database
```

```
db.dropDatabase()
```

4. Output:

```
shivanshuranjansingh@Shivanshu-MacBook ~ % mongosh
Current Mongosh Log ID: 698397ad55d128fed2120e1b
Connecting to: mongos://127.0.0.1:27037/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.6
Using MongoDB: 7.0.25
Using Mongosh: 2.5.6

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2025-10-30T22:06:06.569+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
|
|_ admin          48.00 KiB
|_ config         100.00 KiB
|_ db_shivanshu   60.00 KiB
|_ local          72.00 KiB
test> use db_shivanshu
switched to db db_shivanshu
db_shivanshu> db.createCollection("movies")
{ ok: 1 }
db_shivanshu> |
```

```

db_shivanshu> db.movies.insertOne({
  title: "Inception",
  director: "Christopher Nolan",
  year: 2010,
  rating: 8.8,
  genres: "Sci-Fi",
  features: ["Mind-bending", "Action"],
  added_by: "Shivanshu"
})
{
  acknowledged: true,
  insertedId: ObjectId('698399fe65d128fad2120e1e')
}
db_shivanshu> db.movies.insertMany([
  {
    title: "Interstellar",
    director: "Christopher Nolan",
    year: 2014,
    rating: 8.6,
    genres: "Sci-Fi",
    features: ["Space", "Drama"],
    added_by: "Shivanshu"
  },
  {
    title: "Avatar",
    director: "James Cameron",
    year: 2009,
    rating: 7.8,
    genres: "Adventure",
    features: ["3D", "Action"],
    added_by: "Shivanshu"
  },
  {
    title: "The Dark Knight",
    director: "Christopher Nolan",
    year: 2008,
    rating: 9.0,
    genres: "Action",
    features: ["Batman", "Crime"],
    added_by: "Shivanshu"
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('698399fe65d128fad2120e1d'),
    '1': ObjectId('698399fe65d128fad2120e1e'),
    '2': ObjectId('698399fe65d128fad2120e1f')
  }
}
db_shivanshu>

db_shivanshu> db.movies.find()
[
  {
    _id: ObjectId('698399fe65d128fad2120e1d'),
    title: 'Inception',
    director: 'Christopher Nolan',
    year: 2010,
    rating: 8.8,
    genres: 'Sci-Fi',
    features: [ 'Mind-bending', 'Action' ],
    added_by: 'Shivanshu'
  },
  {
    _id: ObjectId('698399fe65d128fad2120e1e'),
    title: 'Interstellar',
    director: 'Christopher Nolan',
    year: 2014,
    rating: 8.6,
    genres: 'Sci-Fi',
    features: [ 'Space', 'Drama' ],
    added_by: 'Shivanshu'
  },
  {
    _id: ObjectId('698399fe65d128fad2120e1f'),
    title: 'Avatar',
    director: 'James Cameron',
    year: 2009,
    rating: 7.8,
    genres: 'Adventure',
    features: [ '3D', 'Action' ],
    added_by: 'Shivanshu'
  },
  {
    _id: ObjectId('698399fe65d128fad2120e1f'),
    title: 'The Dark Knight',
    director: 'Christopher Nolan',
    year: 2008,
    rating: 9,
    genres: 'Action',
    features: [ 'Batman', 'Crime' ],
    added_by: 'Shivanshu'
  }
]
db_shivanshu> db.movies.findOne()
{
  _id: ObjectId('698399fe65d128fad2120e1e'),
  title: 'Inception',
  director: 'Christopher Nolan',
  year: 2010,
  rating: 8.8,
  genres: 'Sci-Fi',
  features: [ 'Mind-bending', 'Action' ],
  added_by: 'Shivanshu'
}

```

```

db_shivanshu> db.movies.find({}, { title: 1, year: 1, _id: 0 })
[
  { title: 'Inception', year: 2010 },
  { title: 'Interstellar', year: 2014 },
  { title: 'Avatar', year: 2009 },
  { title: 'The Dark Knight', year: 2008 }
]
db_shivanshu> db.movies.find({ genre: "Action" })
[
  {
    _id: ObjectId('698399fe65d128fad2120e1f'),
    title: 'The Dark Knight',
    director: 'Christopher Nolan',
    year: 2008,
    rating: 9,
    genres: 'Action',
    features: [ 'Batman', 'Crime' ],
    added_by: 'Shivanshu'
  }
]
db_shivanshu> db.movies.find({ "details.language": "English" })
[
]
db_shivanshu>

```

```
db_shivanshu> db.movies.updateOne(
  { title: "Avatar" },
  { $set: { rating: 8.1, genre: "Sci-Fi Adventure" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db_shivanshu> db.movies.updateOne(
  { title: "Inception" },
  { $push: { features: "Thriller" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db_shivanshu> db.movies.updateOne(
  { title: "Inception" },
  { $pull: { features: "Action" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db_shivanshu> db.movies.updateMany(
  { director: "Christopher Nolan" },
  { $set: { language: "English" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
db_shivanshu> db.movies.updateMany({}, { $unset: { added_by: "" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
db_shivanshu> db.movies.updateMany({}, { $set: { color: "Full HD" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
db_shivanshu> db.movies.updateOne(
  { title: "Tenet" },
  { $set: { director: "Christopher Nolan", year: 2020, rating: 7.6 } },
  { upsert: true }
)
{
  acknowledged: true,
  insertedId: ObjectId('49037ac8e6c81ac5b85a7e326'),
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 1
}
db_shivanshu> █
```

```
db_shivanshu> db.movies.deleteOne({ title: "Avatar" })
{ acknowledged: true, deletedCount: 1 }
db_shivanshu> db.movies.deleteMany({ director: "Christopher Nolan" })
{ acknowledged: true, deletedCount: 4 }
db_shivanshu> db.movies.deleteMany({})
{ acknowledged: true, deletedCount: 0 }
db_shivanshu> █
```

```
db_shivanshu> db.movies.aggregate([
|   { $group: { _id: "$director", total_movies: { $sum: 1 } } }
| ])
[]

db_shivanshu> db.movies.aggregate([
|   { $group: { _id: "$genre", total_movies: { $sum: 1 } } }
| ])
[]

db_shivanshu> 
```

```
db_shivanshu> db.movies.drop()
[]
true
db_shivanshu> db.dropDatabase()
[]
{ ok: 1, dropped: 'db_shivanshu' }
db_shivanshu> 
```

5. Learning Outcomes:

- Understand the structure and working of MongoDB as a NoSQL database. □ Create and manage databases and collections using MongoDB commands.
- Perform CRUD (Create, Read, Update, Delete) operations on documents.
- Apply aggregation operations for data grouping and analysis.
- Gain practical skills in handling and modifying data using MongoDB shell.